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No Undeclared Boundary: Scope, Approximation, and Admissibility in Relation-First Inference

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Abstract

The Relation-First Organization Programme currently has a mature master no-go principle:

$$\boxed{\text{No declared relation, no legitimate relational inference.}}$$

This paper derives and formalizes the boundary face of that principle.

The result is not a new axiom competing with No Undeclared Relation. It is a necessary consequence of taking relation as primitive.

A relation cannot be declared unless the relata, distinction classes, admissibility limits, support conditions, or comparison boundaries that make the relation determinate are also declared.

Thus:

$$\boxed{\text{No declared boundary, no declared distinction.}}$$

is a corollary face of:

$$\boxed{\text{No Undeclared Relation.}}$$

The paper proves that every distinction claim, exactness claim, approximation claim, support claim, failure claim, local/global claim, and transfer claim requires a declared boundary. Without such a boundary, the claim is not merely imprecise; it is an under-declared relation.

The central theorem is the **Boundary-Face Necessity Theorem**:

$$\boxed{\text{If relation is primitive, then declared relation requires declared boundary.}}$$

This paper draws on the boundary machinery of the Theory of Organization, the boundary recurrence machinery of Triadic Closure, and the calibrator-boundedness machinery of Recursive Bridge Calibration.

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0. Scope

This paper is a foundational no-go paper parallel in status to **No Undeclared Relation and Bridge Completion**, but subordinate in logical order.

It does not replace No Undeclared Relation.

It derives one face of it.

The hierarchy is:

$$\boxed{\text{relation-first primitive} \Rightarrow \text{No Undeclared Relation} \Rightarrow \text{No Undeclared Boundary.}}$$

This paper does not claim:

1. that boundary is a separate primitive competing with relation;

2. that all boundaries are spatial;
3. that all distinctions are sharp;
4. that all boundaries are physical;
5. that every exact result is globally total;
6. that approximation is vague.

It claims:

a relation cannot be declared without declaring the boundary that makes its distinctions determi

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1. Motivation

A relation-first programme cannot treat boundaries as optional.

To claim:

$$xRy$$

one must know what is related, what counts as x, what counts as y, what distinguishes related from unrelated, and what limits the claim.

Otherwise the relation is not declared.

Examples:

proper organization

requires the boundary between proper interior and degenerate extremes.

persistence support

requires the boundary of threshold support.

approximation

requires tolerance.

exactness

requires the local target boundary at which error vanishes.

failure

requires the boundary crossed or violated.

$$\text{objecthood}$$

requires boundaries of charted compatibility.

$$\text{count-volume calibration}$$

requires the tolerance boundary between admissible and failed volume comparison.

Thus the boundary is not decorative. It is what lets a relation claim have determinate content.

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2. Relation-first derivation of boundary

2.1 Relation requires relata

A relation is not a relation between nothing.

If:

$$R$$

is declared, then some distinction must make available what is related.

This does not require object-first primitives. It requires boundary faces internal to the relation.

The boundary answers:

$$\boxed{\text{what is being distinguished so that relation can be asserted?}}$$

2.2 Boundary is not objecthood

Declaring a boundary does not mean declaring an object.

A boundary may be:

- a threshold;
- a support edge;
- a zero cut;
- an admissibility condition;
- a chart domain;
- an interval endpoint;
- an error tolerance;
- a local task boundary;
- a proper/improper distinction;

- a source/target distinction;
- an initial/terminal limit.

The boundary is a relation-face, not necessarily an object.

2.3 Boundary is forced by relation

If a supposed relation has no boundary at all, then it has no determinate distinction between:

- source and target;
- inside and outside;
- admissible and inadmissible;
- exact and approximate;
- success and failure;
- proper and degenerate;
- local and global.

Such a “relation” cannot be inspected or tested.

Therefore boundary declaration is required by relation declaration.

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3. Internal sources in the programme

3.1 Theory of Organization

The Theory of Organization supplies the strongest boundary machinery:

- answer face;
- question face;
- zero-boundary face;
- proper organization layer;
- excluded boundary extremes;
- persistence support;
- threshold cuts;
- evaluation over proper organizations.

A proper organization exists only inside a boundary discipline. It is not at the collapsed initial extreme and not at terminal saturation.

3.2 Triadic Closure

Triadic Closure explicitly treats boundary as one of the required modes of relation-first closure:

$$\boxed{\text{boundary} - \text{mediation} - \text{visibility.}}$$

Boundary recurrence:

$$z_- \equiv_\partial z_+$$

does not collapse the proper interior. It shows that initial and terminal boundary extremes are not themselves the proper relational state.

3.3 Recursive Bridge Calibration

Recursive Bridge Calibration already uses boundary through:

$$\mathfrak{K} = (D'_Y, \varepsilon, \mathcal{E}, \mathrm{Adm}, \mathrm{Update}).$$

Here:

- D'_Y declares the target boundary;
- ε declares tolerance;
- Adm declares success/failure boundary;
- $z_- < \mathfrak{K} < z_+$ declares calibrator boundedness.

Thus approximation and exactness are already boundary-governed.

3.4 Density / anchoring branch

The recent density and anchoring papers rely on boundary:

- interval classes;
- internal density classes;
- anchoring error;
- tolerance;
- faithful interval-region anchoring;
- canonical region target;
- Lorentzian realization obstruction.

The GR-facing branch therefore depends on boundary discipline.

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4. Core definitions

Definition 4.1 — Boundary face

A **boundary face** of a relation claim is any declared structure that separates, limits, supports, admits, excludes, localizes, or compares the distinction classes used by that relation.

It may be sharp, fuzzy, metric, order-theoretic, topological, categorical, logical, operational, empirical, or task-relative.

Definition 4.2 — Boundary declaration

A **boundary declaration** is a tuple:

$$\mathfrak{B}_\partial = (D, \partial D, \mathrm{Adm}, \varepsilon, \mathrm{Fail})$$

where:

1. D is the distinction family;
2. ∂D is the boundary or support condition;
3. Adm is the admissibility rule;
4. ε is tolerance, if approximation is involved;
5. Fail records boundary failure modes.

Not every application needs all five components explicitly. But whatever component does boundary work must be declared.

Definition 4.3 — Underdeclared boundary

A boundary is **underdeclared** when a claim uses a distinction, support, exactness, tolerance, admission, failure, or local/global status without declaring the boundary by which that status is determined.

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5. Main theorem

Theorem 5.1 — Boundary-Face Necessity Theorem

If relation is primitive, then every declared relation requires a declared boundary face sufficient to determine the distinctions used by that relation.

Proof

Let R be a claimed relation. To declare R, one must determine what the relation relates and what counts as success or failure of the relation claim. This requires a distinction between at least some of:

- related / unrelated;
- source / target;
- admissible / inadmissible;
- inside / outside;
- support / exterior;
- exact / non-exact;
- approximate / non-admissible.

Such distinctions require boundary conditions. If the boundary conditions are not declared, then the relation's domain, target, admissibility, or success condition is underdeclared. Therefore R is itself underdeclared.

Thus boundary declaration is necessary for declared relation. □

Corollary 5.2 — No Undeclared Boundary

No distinction-bearing inference is legitimate unless the boundary supporting the relevant distinction is declared.

Proof

By Theorem 5.1, a relation using a distinction requires the boundary making that distinction determinate. If the boundary is undeclared, the relation is underdeclared and violates No Undeclared Relation. □

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6. Single-face instability

The primitive role set is:

$$\{\mathsf{R}, \mathsf{O}, \mathsf{Z}\}.$$

No single face is stable alone.

6.1 Boundary alone: $\{\mathsf{Z}\}$

The singleton boundary face:

$$\{\mathsf{Z}\}$$

is inert boundary.

It marks a limit, cut, threshold, support, or endpoint, but supplies neither mediation nor visibility.

Thus boundary alone cannot declare relation.

6.2 Mediation alone: $\{\mathsf{R}\}$

The singleton mediation face:

$$\{\mathsf{R}\}$$

is undisciplined connectivity.

It supplies connection or transition without declaring what is bounded or how the connection is read.

6.3 Observer/readout alone: $\{\mathsf{O}\}$

The singleton observer/readout face:

$$\{\mathsf{O}\}$$

is arbitrary perspective.

It supplies visibility or readout without declaring what boundary is being read or what mediation grounds the readout.

6.4 Boundary-face lesson

No Undeclared Boundary is not the claim that boundary is sufficient.

It is the claim that boundary is necessary as one face of declared relation.

A bare boundary is unstable. A relation using a boundary must still declare mediation and observer/readout.

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7. Boundary in triadic substitution and coupling

The boundary face is not isolated. It is a role in a triadic declaration.

By the Relation-to-Triad Declaration theorem:

$$R_{\text{declared}} \Rightarrow (\mathsf{R}, \mathsf{O}, \mathsf{Z}).$$

No Undeclared Boundary is the Z-face of No Undeclared Relation.

7.1 Boundary substitute

A **boundary substitute** is any structure used to perform the Z-role in a local triad:

- threshold;
- support;
- margin;
- tolerance;
- endpoint;
- domain;
- proper/interior cut;
- local/global boundary;
- exactness boundary.

Theorem 7.2 — Boundary Substitution Theorem

A nontrivial boundary substitute is admissible only if it either satisfies the boundary role locally or carries a declared triadic closure sufficient for that role.

Proof

This is the Triadic Substitution Principle applied to the Z-role. If the boundary substitute does nontrivial relation work, it must have declared mediation, visibility, and boundary conditions sufficient for that work. Otherwise the boundary substitute is underdeclared. □

7.3 Boundary coupling

When a boundary from one triad is used in another triad, it forms a role interface.

The interface is admissible only if it lands in a closure-complete triadic context.

Proposition 7.4 — No Dangling Boundary Interface

A boundary face inserted into another relation is admissible only if it lands in a declared closure-complete context.

Proof

A boundary face used in another relation is a role interface. By the No Dangling Role Interface proposition, a role-face is admissible only when it lands in a closure-complete triadic context. Therefore a boundary cannot hang free as an unpaired threshold, support, or tolerance. □

7.5 Boundary and unstable dyads

Boundary appears in two dyadic unstable configurations:

$$\{\mathsf{Z}, \mathsf{R}\}$$

boundary + mediation without visibility;

$$\{\mathsf{Z}, \mathsf{O}\}$$

boundary + visibility without mediation.

Thus a declared boundary alone does not stabilize a relation. It must be coupled to mediation and readout.

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8. Distinction, exactness, and approximation

Theorem 8.1 — No Distinction Without Boundary

If a claim distinguishes A from B, then it presupposes a boundary separating the distinction classes. If that boundary is undeclared, the distinction is underdeclared.

Proof

A distinction A/B requires some condition under which a case is admitted as A, admitted as B, both, neither, approximate, or indeterminate. That condition is the boundary face. Without it, the distinction has no declared criterion. □

Theorem 8.2 — Approximation Requires Boundary

A claim of approximate recovery is admissible only if a tolerance or admissibility boundary is declared.

Proof

Approximation means that a source result is close enough to a target for the declared purpose. "Close enough" is meaningless without a tolerance, error rule, comparison class, or admissibility boundary. Therefore approximation requires boundary declaration. □

Theorem 8.3 — Exact Local Calibration as Boundary Contact

If a result is exact relative to a declared local target, then it reaches the terminal boundary of that local calibrated task.

Proof

Let a calibrated target be:

$$(D'_Y, \varepsilon, \mathfrak{K}).$$

Exactness means:

$$\varepsilon = 0$$

and the recovered distinctions match D'_Y. Therefore the calibration error is zero relative to that local target. This is terminal contact for the local task boundary. It does not imply global completion unless the target itself is global and total. □

Theorem 8.4 — Global Exactness Requires Declared Global Boundary

If a claim asserts exact recovery of all possible target distinctions, then it asserts terminal totalization relative to the declared target. Such a claim is admissible only if the global target boundary is declared and the proof accounts for it.

Proof

All possible target distinctions constitute the full target boundary. Exact recovery of all of them leaves no unrecovered remainder. That is a terminal-boundary claim relative to the target. By No Undeclared Boundary, the target boundary must be declared; by No Undeclared Relation, transfer across that boundary is not free. □

Remark 8.5 — Asymptotic or unreachable global boundaries

Global exactness is not automatically forbidden. What is forbidden is undeclared globality.

In physics, a result may be exact relative to a declared global or asymptotic boundary condition, for example:

- asymptotic flatness;
- boundary conditions at infinity;
- compact support;
- fixed variation class;
- prescribed falloff conditions.

These are not counterexamples to No Undeclared Boundary. They are examples of boundary discipline.

The correct distinction is:

$$\boxed{\text{global exactness with declared asymptotic/global boundary} \neq \text{undeclared totalization.}}$$

The first is admissible if proved. The second is overclaim.

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9. Boundary no-go family

The following are summary projections of the theorems above.

No-go	Boundary being demanded	Source theorem
no undeclared threshold	admissible / inadmissible cut	Theorem 8.1
no undeclared support	support / exterior boundary	Theorem 8.1
no		

The key slogan is: $\boxed{\text{exact for } D'_Y \neq \text{exact for all } D_Y.}$

10. Boundary-face recursion

Boundary is the Z-face of declared relation.

In the triadic recursion decomposition, boundary corresponds to residual recursion:

$$\boxed{\text{Z-recursion} = \text{residual recursion.}}$$

10.1 Residual as boundary remainder

A proper nonterminal closure is neither lower-boundary collapse nor terminal totalization. It leaves a residual:

$$\Omega \neq 0.$$

This residual is a boundary remainder: the local closure has not vanished relative to the declared target boundary.

10.2 Boundary pressure and realized recursion

Residual boundary pressure is not yet realized recursion. It becomes realized recursion only when an admitted update or trace uses the residual to change later available structure.

Thus:

$$\boxed{\text{boundary residual} \to \text{recursion pressure} \to \text{admitted update} \to \text{realized recursion.}}$$

10.3 Boundary role in closure-atlas pathing

In a triadic closure atlas, an unresolved boundary term is not just an error. It is a pathing signal. It identifies the next closure obligation.

This is why exactness, approximation, persistence support, density anchoring, and GR continuum calibration all depend on declared boundary structure.

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11. Examples across the programme

11.1 Proper organization

A proper organization is not merely a fixed point. It is a fixed point in the proper interior, excluding boundary extremes.

No declared proper boundary, no proper organization.

11.2 Persistence support

A persistence support:

$$I = (\ell, u]$$

is meaningful only because ℓ and u declare boundary.

No support boundary, no persistence claim.

11.3 Moving supports

In moving-support trace stability, the combined budget:

$$|\delta| + \eta < m$$

works because the boundary margin m is declared.

No margin boundary, no stability claim.

11.4 Noether-style conservation

A Noether-style conservation theorem requires declared boundary data.

The relevant boundary faces include:

- action domain;
- admissible variation class;
- boundary terms;
- symmetry class;
- support/falloff assumptions.

Without those, the conservation claim is underdeclared.

The conservation law may be exact, but it is exact relative to the declared action, variation class, and boundary conditions. This is local/global boundary discipline, not totalization.

11.5 Effective field theory scope

Effective field theory is a canonical physics example of boundary discipline.

An EFT result is not a claim about all scales. It is a claim inside a declared regime:

- energy scale;
- cutoff;
- degrees of freedom;
- relevant/irrelevant operators;
- matching conditions;
- error terms.

The EFT boundary is what makes the result exact or approximate in the intended regime.

No declared regime, no legitimate EFT inference.

11.6 Asymptotic flatness and boundary at infinity

In general relativity, global quantities such as ADM mass are defined only under declared asymptotic boundary conditions.

The boundary may be "at infinity" and not reachable as a local finite endpoint. But it is still declared.

Thus asymptotic boundary conditions illustrate the correct reading of Theorem 8.4:

$$\boxed{\text{unreachable boundary can be declared;}}$$

$$\boxed{\text{undeclared globality cannot.}}$$

11.7 Count-volume calibration

The anchoring error:

$$E_{\text{anchor}}(U) = |\alpha\mu_{\text{int}}(U) - \text{Vol}_M(\Psi(U))|$$

requires a tolerance boundary. Otherwise no admissibility claim can be made.

11.8 GR-facing bridge

The distinction between:

$$\mathcal{R}_{\text{can}}$$

and:

$$\mathcal{R}_M$$

is a boundary distinction. Without declaring the boundary between abstract canonical region space and Lorentzian continuum target, the GR bridge is overclaimed.

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12. Relation to No Undeclared Relation

No Undeclared Boundary is not a rival principle.

It is a corollary face.

$$\boxed{\text{undeclared boundary} \Rightarrow \text{underdeclared relation} \Rightarrow \text{NUR failure.}}$$

NUR blocks the invisible arrow.

No Undeclared Boundary identifies one way the arrow remains invisible: the distinction-supporting boundary has not been declared.

Thus the master principle remains:

$$\boxed{\text{No Undeclared Relation.}}$$

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13. Relation to No Undeclared Observer

Boundary and observer/readout are distinct faces.

A boundary may be declared but unreadable.

An observer/readout may exist but lack a boundary of admissibility.

A calibrated relation needs both:

$$\boxed{\text{boundary} + \text{observer/readout.}}$$

Thus No Undeclared Boundary and No Undeclared Observer are sibling consequences of relation-first, not independent primitives.

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14. Claims and non-claims

14.1 Claims

This paper claims:

1. boundary declaration is required by relation-first;
2. No Undeclared Boundary is a corollary face of No Undeclared Relation;
3. boundary alone is one of the three unstable singleton configurations;
4. every distinction-bearing claim requires a boundary;
5. approximation requires tolerance boundary;
6. exactness is local to a declared target boundary;
7. global exactness requires a declared global or asymptotic boundary;
8. undeclared global exactness is terminal-totalization overclaim;
9. nontrivial boundary substitutes require triadic declaration;
10. dangling boundary interfaces are underdeclared;
11. many programme no-gos are boundary-face instances of NUR.

14.2 Non-claims

This paper does not claim:

1. boundary is an independent primitive over relation;
2. every boundary is spatial or physical;
3. boundary alone is sufficient for relation;
4. every distinction is crisp;
5. local exactness implies global completion;
6. every boundary can be easily formalized;
7. global exactness is always forbidden;
8. No Undeclared Boundary replaces No Undeclared Relation;
9. a boundary substitute is valid without triadic declaration.

15. Summary

If relation is primitive, boundary is required.

A relation between nothing is no relation.

A distinction without boundary is underdeclared.

Approximation without tolerance is underdeclared.

Exactness without target boundary is underdeclared.

Globality without global boundary is overclaim.

Therefore:

$$\boxed{\text{No declared boundary, no declared distinction.}}$$

But this is not a new master axiom. It is the boundary face of:

$\boxed{\text{No Undeclared Relation.}}$

Internal references

- Theory of Organization.
- Triadic Closure and Recursive Residual Theory.
- Recursive Bridge Calibration.
- No Undeclared Relation and Bridge Completion.
- Density, Sampling, and Faithful Embedding.

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